

Reframing Human–AI Interaction through Tangible, Multisensory Interfaces

Suk Min Hwang
University of California, Berkeley
Berkeley, CA, USA
sukmin@berkeley.edu

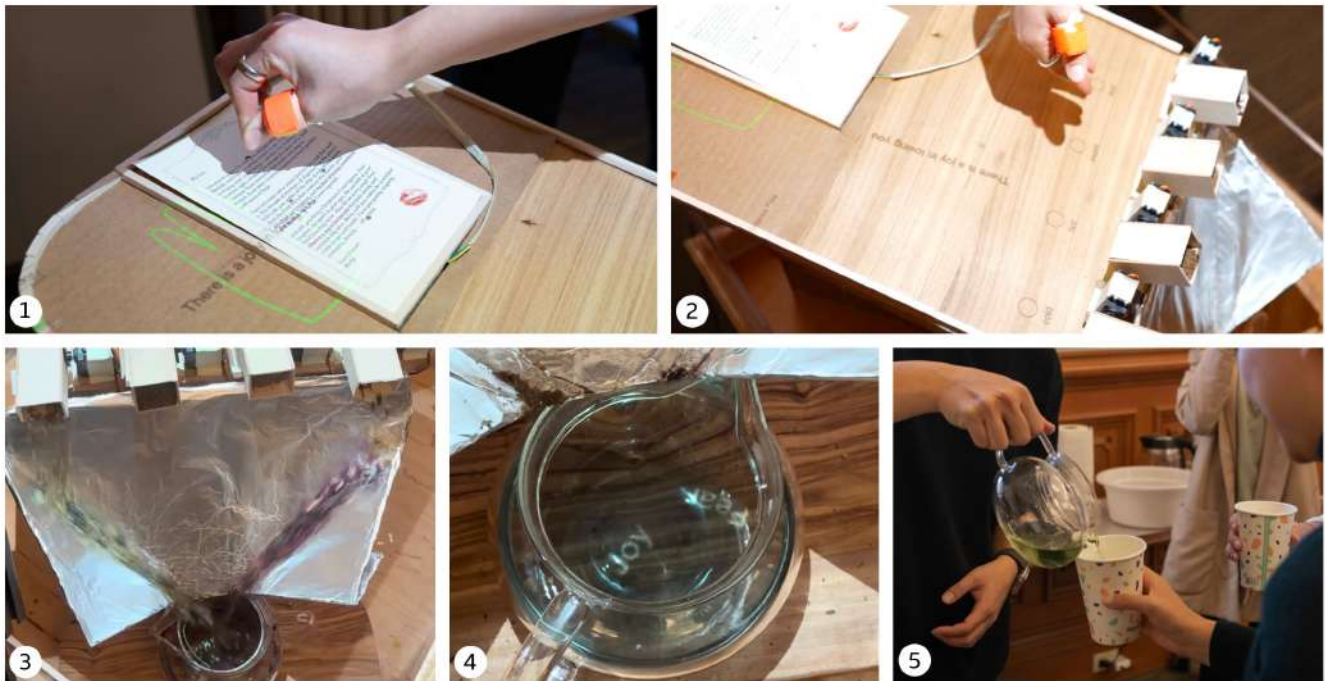


Figure 1: Interaction sequence: (1) a phrase is physically picked up, (2) the phrase animates toward the teapot, (3) mapped tea leaves are dispensed, (4) emotional qualities surface during brewing, and (5) the tea is consumed individually or shared socially.

Abstract

We present a tangible, multisensory AI system that translates personal text into tea. Users select phrases through pinch gestures and release them into a teapot; as each phrase flows down, a large language model classifies its emotional tone and triggers the corresponding tea ingredients. By combining multiple phrases, users produce layered flavor compositions that reflect their curatorial choices. Observations from a public deployment with approximately 40 participants suggest that this embodied framing shifted how people engaged with AI interpretation: rather than evaluating correctness, participants treated the AI's mappings as creative constraints, developed a sense of authorship through physical curation and consumption, and explored meaning through taste and social sharing.

We discuss design considerations for AI systems that foreground user agency over accuracy-driven evaluation.

ACM Reference Format:

Suk Min Hwang. 2026. Reframing Human–AI Interaction through Tangible, Multisensory Interfaces. In *Extended Abstracts of the 2026 CHI Conference on Human Factors in Computing Systems (CHI EA '26)*, April 13–17, 2026, Barcelona, Spain. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3772363.3798597>

1 Introduction

As AI systems increasingly interpret human language for emotional meaning, interaction design shapes how users engage with these outputs. Most current systems present AI interpretations as definitive labels, reinforcing a dynamic where the user's role is to accept or reject [12, 14]. Yet emotion and memory resist fixed categorization; they are products of situated, embodied experience [4, 8].

Tangible User Interfaces (TUIs) offer one way to reframe this dynamic by giving data material form [9, 10]. We argue that externalizing AI interpretation into an edible, multisensory form can



This work is licensed under a Creative Commons Attribution 4.0 International License. *CHI EA '26, Barcelona, Spain*

© 2026 Copyright held by the owner/author(s).

ACM ISBN 979-8-4007-2281-3/26/04

<https://doi.org/10.1145/3772363.3798597>

invite what we call *material claiming*—the act of physically selecting, shaping, and consuming an output that makes it feel like one’s own. In this framing, the AI provides sensory “ingredients”; the user authors the experience.

To explore this, we designed an interface that translates AI interpretations of personal letters into tea. Users select phrases through a pinch gesture and release them toward a teapot; a large language model (GPT-4) classifies each phrase’s emotional tone and triggers corresponding tea ingredients. Multiple phrases can be combined, producing layered flavors. Tea was chosen deliberately: it carries associations with slowness, presence, and ritual, and unlike screen-based outputs, a cup of tea must be held, smelled, tasted, and can be shared. Through observations from a public deployment with approximately 40 participants, we characterize how this approach encouraged participants to treat AI outputs as creative materials rather than fixed labels, and offer design considerations for embodied AI systems that prioritize exploration over evaluation.

2 Related Work

Tangible Interaction and Human-AI Collaboration. Tangible User Interfaces (TUIs) give data material form, enabling users to construct meaning through physical manipulation rather than passive reception [4, 8, 9]. In Human-AI Interaction, a parallel shift is underway: traditional XAI approaches help users judge AI correctness [1, 14], but this positions AI as an oracle and the user as its evaluator [12, 16]. Recent work reframes AI as a co-creative partner that provides generative constraints rather than definitive answers [5, 7, 11]. However, most co-creative AI systems still operate through screens. This work asks what happens when AI interpretation takes the form of a tangible, edible artifact that users physically curate and consume.

Multisensory and Edible Interfaces. Unlike vision and language, which are processed through cortical pathways, smell and taste connect directly to the amygdala and hippocampus—brain regions central to emotion and memory [13, 17]. This makes edible media a compelling but underexplored channel for human-AI interaction. Prior Food-HCI systems have translated data into edible forms—turning journaling data into hot cocoa [6] or laughter into chocolate [15]—but have largely focused on data visualization or sensory novelty rather than on reframing how users relate to AI interpretation. We explore this gap: by manifesting AI-mediated interpretation as tea, the act of consuming an output makes the experience irreversibly personal and invites social sharing in ways that screen-based interfaces cannot.

3 System

We designed a system that translates AI-mediated interpretation of personal letters into an edible, tea-based experience through physical interaction, LLM-based emotion classification, and custom sensing hardware.

3.1 Interaction Flow

The interaction follows four stages. First, participants choose from three physical letters representing different emotional contexts: a love letter, a job offer, and an inspirational message. Each letter

contains three paragraphs anchored to distinct emotional tones. Second, participants physically select phrases using a pinch gesture—pressing thumb and index finger together as if picking up a tangible object (Figure 1). A projector displays the selected text around the participant’s hand. The text follows the hand’s movement and scales with proximity, creating the sensation of physically holding and moving the phrase. Third, when participants release the pinch, the selected phrase visually flows down a tilted surface toward four tea dispensers. Based on the phrase’s emotional mapping, specific dispensers activate and release tea ingredients into a teapot. Participants can select multiple phrases in sequence, each adding layers to the flavor profile. Emotional keywords appear as floating text projected onto the teapot, making the AI’s interpretation partially visible. Finally, participants receive their brewed tea and are invited to taste and reflect on how the flavor embodies their choices.

3.2 LLM-Based Emotion-to-Flavor Translation

The system translates selected phrases into tea flavors through AI-mediated classification. When a participant selects a phrase, it is sent to a large language model (GPT-4, OpenAI) via API with a structured prompt. The prompt provides seven predefined emotion categories and instructs the model to classify the input phrase into one of these categories. The model returns a single emotion label per phrase.

The emotion categories were developed through an iterative design process. The initial design began with five basic emotions—joy, anger, fear, anxiety, and sadness—inspired by widely recognized models of discrete affect. Through refinement based on the emotional range present in the source letters, sadness was removed and three composite emotions were introduced: passion (anger + joy), nostalgia (fear + joy), and disappointment (fear + anxiety). These composite categories better captured the layered affective tones of personal correspondence and mapped structurally onto blended tea flavors.

Each emotion is associated with a specific flavor profile and tea ingredient (Table 1). Four base emotions map to individual flavors: joy to sweet (cocoa berry), anxiety to sour (hibiscus), anger to hot/spicy (spiced chai), and fear to cold/minty (peppermint with butterfly pea). Three composite emotions combine these base flavors: passion (hot + sweet), nostalgia (cold + sweet), and disappointment (cold + sour). The mapping draws on both crossmodal taste-emotion

Table 1: Emotion to flavor and tea ingredient mapping used in the system.

Emotion	Flavor Profile	Tea Ingredient
Joy	Sweet	Cocoa berry
Anxiety	Sour	Hibiscus
Anger	Hot (spicy)	Spiced chai
Fear	Cold (minty)	Peppermint + butterfly pea
Passion	Hot + Sweet	Spiced chai + cocoa berry
Nostalgia	Cold + Sweet	Peppermint + butterfly pea + cocoa berry
Disappointment	Cold + Sour	Peppermint + butterfly pea + hibiscus

research [13, 17] and the physiological properties of tea ingredients themselves—for instance, peppermint produces a cooling sensation that aligns with the withdrawal associated with fear, while spiced chai generates warmth that parallels the intensity of anger. When participants select multiple phrases, the corresponding ingredients combine during brewing, producing a layered flavor that reflects the emotional content of their selections. The goal of these mappings is not correctness but legibility—an exploratory translation that invites reflection rather than verification.

3.3 Architecture

The system integrates sensing, projection, computation, and actuation into a unified pipeline. Phrase selection is detected using a force-sensitive resistor (FSR) for pinch detection and an array of photocells for spatial sensing (Figure 2). Tea ingredients are dispensed via four wooden containers, each mounted on a servo motor controlled by an Arduino UNO R4. An overhead projector provides real-time visual feedback. A p5.js-based projection layer visualizes phrase selection and movement. Selected phrases are sent to GPT-4 for classification, and the system triggers servo actuation based on the returned mappings.

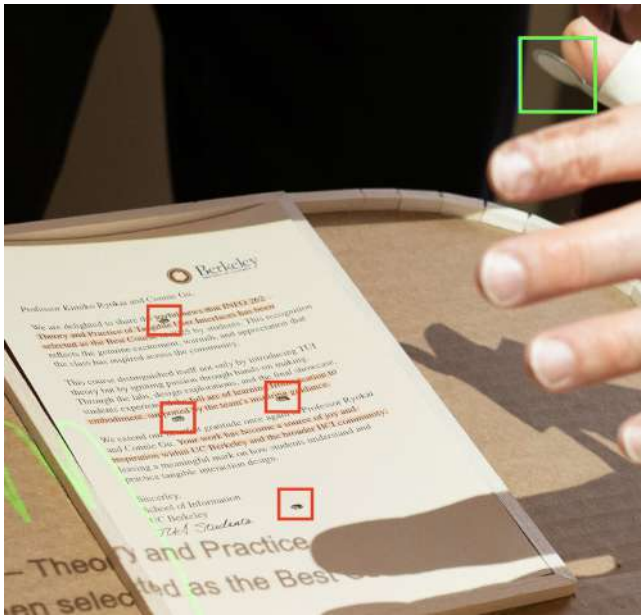


Figure 2: Close-up of the sensing hardware embedded in the letter surface. Red boxes indicate photocells used for spatial sensing of hand position; the green highlight marks the force-sensitive resistor (FSR) that detects the pinch gesture.

4 Discussion

4.1 Deployment Context

The system was deployed at a two-day public showcase at South Hall, UC Berkeley, where approximately 40 participants (primarily students, faculty, and staff) interacted with the installation. Data was collected through field notes and informal conversations with

participants. No formal interview protocol or video recording was used. The observations below reflect recurring patterns rather than systematic qualitative analysis, and all claims should be interpreted within the scope of this preliminary evidence.

4.2 Observed Patterns

Four patterns emerged from the deployment.

Playful exploration through tangible word interaction. The physical act of pinching to “pick up” projected text created a visceral sense of grasping words. Participants reacted with surprise and delight, with several exclaiming that they were “actually holding words.” This playfulness invited bystanders to join and motivated participants to explore beyond a single phrase—selecting multiple sentences to discover how different combinations would translate, creating an iterative loop rather than a one-off reading. Among the three letter types, the job offer letter was most popular, with the phrase “We’re delighted to offer you a position” being selected most frequently. This suggests that embodied gestures can encourage open-ended, curiosity-driven engagement with AI outputs.

A sense of authorship through embodied curation. The pinch-and-release gesture mirrored adding ingredients while cooking. Participants selected their source material, physically shaped the brewing process, and consumed the result. This end-to-end involvement appeared to cultivate ownership: many photographed their tea or held the cup carefully, treating it as a personal creation. Several returned to try different combinations. Just as hand-selecting ingredients makes a dish feel like “mine,” curating phrases into tea seemed to foster authorship over the interpretation—suggesting that end-to-end physical curation, from selection through consumption, may shift users from judging outputs to engaging with them as personal creations.

Multisensory social sharing. Participants experienced text through color, aroma, temperature, and taste, transforming reading into a full-bodied sensory encounter. When a phrase mapped to “joy” became cocoa berry tea, participants could see its warm color, smell its sweetness, and taste its richness. This translation invited social sharing: one group brewing tea from the job offer letter described it as “celebrating this achievement together,” noting that drinking collectively felt like sharing the letter’s emotions in a way reading aloud could not (Figure 3). Materializing outputs through multiple senses can transform interpretation from a solitary task into a shared experience.

Exploratory rather than evaluative inquiry. Participants’ questions about the system tended to be exploratory rather than evaluative. Rather than asking “Did the AI get it right?,” they asked “What does this phrase give me?” or “What happens if I combine joy and anxiety?” This suggests participants treated the AI’s mappings as creative constraints to explore rather than verdicts to verify. At the same time, participants frequently asked how the emotion-to-flavor mapping was determined—the most common question across the deployment—indicating genuine curiosity about the semantic-to-sensory translation logic rather than skepticism toward it. Exposing mappings as explorable constraints, rather than hiding them, may be key to encouraging this curiosity-driven stance.



Figure 3: Participants sharing tea brewed from selected letter phrases during the public showcase at South Hall, UC Berkeley.

4.3 Limitations and Future Work

This work presents an exploratory system with preliminary observations. The findings are based on naturalistic observation and informal conversations rather than a formal user study, and the sample was limited to a university population. A controlled study with video analysis, think-aloud protocols, and semi-structured interviews would provide more systematic insights into how participants experience authorship over AI-mediated interpretation and how different design choices shape this experience.

Several technical limitations constrain the current system. The use of photocells and a force-sensitive resistor occasionally caused the same phrase to be selected repeatedly, limiting participants' ability to explore diverse combinations. The current system also uses physical letters as source material, constraining contextual diversity. Future iterations could integrate computer-vision-based gesture recognition for improved precision, support free-form text selection, and expand to diverse document types—journals, postcards, diplomas—each carrying distinct affective qualities.

The emotion-to-flavor mapping, while grounded in crossmodal research [13, 17], remains inherently subjective and does not account for individual taste preferences or cultural variation. Cross-cultural variation in taste-emotion associations is underexplored; emerging evidence suggests some crossmodal perceptions may be more consistent across cultures than previously assumed [2], but this warrants investigation in the context of edible interfaces. Recent advances in computational taste interfaces [3] suggest possibilities for more fine-grained, real-time flavor modulation beyond pre-mapped ingredients, which could enable personalized or adaptive mappings in future systems.

Beyond tea, future work could explore other co-creative outputs where tangible AI interpretation fosters authorship, such as personalized food, cocktails, or scent compositions. Comparative studies across sensory modalities (taste, smell, haptics, sound) could deepen understanding of how different sensory channels shape human-AI collaboration.

5 Conclusion

This work explored how externalizing AI interpretation as an edible experience can reframe the user's role from evaluator to author. By translating personal text into tea through physical curation, the system turned interpretation into a shared, bodily encounter—one that participants treated with ownership, curiosity, and care. While these observations are preliminary, they suggest that when AI outputs can be held, tasted, and shared, new modes of human-AI collaboration become possible: ones grounded in exploration and affect rather than accuracy alone.

Acknowledgments

To Professor Kimiko Ryokai for her guidance and insightful feedback throughout this project, and to Apolline Tardy, Jasmine Shih, Zhixuan Wang, and Kristen Larson for their thoughtful input and contributions during the development of this work.

References

- [1] Saleema Amershi, Dan Weld, Mihaela Vorvoreanu, Adam Fourney, Besmira Nushi, Penny Collisson, Jina Suh, Shamsi Iqbal, Paul N. Bennett, Kori Inkpen, Jaime Teevan, Ruth Kikin-Gil, and Eric Horvitz. 2019. Guidelines for Human-AI Interaction. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [2] Artin Arshamian, Richard C. Gerkin, Nicole Kruspe, Ewelina Wnuk, Simeon Floyd, Carolyn O'Meara, Gabriela Garrido Rodriguez, Johan N. Lundström, Joel D. Mainland, and Asifa Majid. 2022. The Perception of Odor Pleasantness Is Shared across Cultures. *Current Biology* 32, 9 (2022), 2061–2066.e3. doi:10.1016/j.cub.2022.02.062
- [3] Jas Brooks, Noor Amin, and Pedro Lopes. 2023. Taste Retargeting via Chemical Taste Modulators. In *Proceedings of the 36th Annual ACM Symposium on User Interface Software and Technology*. 1–15. doi:10.1145/3586183.3606818
- [4] Paul Dourish. 2001. *Where the Action Is: The Foundations of Embodied Interaction*. MIT Press.
- [5] Graham Dove, Kim Halskov, Jodi Forlizzi, and John Zimmerman. 2017. UX Design Innovation: Challenges for Working with Machine Learning as a Design Material. In *Proceedings of the 2017 CHI Conference on Human Factors in Computing Systems*. 278–288.
- [6] Keming Gao, Joanne Ma, Julia Wang, and Clarissa Wu. 2022. "How Much Connection Did You Get?" Drinking Journaling Data through Hot Cocoa. In *Extended Abstracts of the 2022 CHI Conference on Human Factors in Computing Systems*. 1–6. doi:10.1145/3491101.3519800
- [7] William W. Gaver, Jacob Beaver, and Steve Benford. 2003. Ambiguity as a Resource for Design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 233–240.
- [8] Eva Hornecker and Jacob Buur. 2006. Getting a Grip on Tangible Interaction: A Framework on Physical Space and Social Interaction. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 437–446.
- [9] Hiroshi Ishii and Brygg Ullmer. 1997. Tangible Bits: Towards Seamless Interfaces between People, Bits and Atoms. In *Proceedings of the ACM SIGCHI Conference on Human Factors in Computing Systems*. 234–241.
- [10] Yvonne Jansen, Pierre Dragicevic, Petra Isenberg, Jason Alexander, Abhijit Karnik, Johan Kildal, Sriram Subramanian, and Kasper Hornbæk. 2015. Opportunities and Challenges for Data Physicalization. In *Proceedings of the 33rd Annual ACM Conference on Human Factors in Computing Systems*. 3227–3236.
- [11] Pegah Karimi, Jeba Rezwana, Safat Siddiqui, Mary Lou Maher, and Nasrin Dehbozorgi. 2020. Creative Sketching Partner: An Analysis of Human-AI Co-Creativity. In *Proceedings of the 25th International Conference on Intelligent User Interfaces*. 221–230.
- [12] Todd Kulesza, Simone Stumpf, Margaret Burnett, Sherry Yang, Irwin Kwan, and Weng-Keen Wong. 2013. Too Much, Too Little, or Just Right? Ways Explanations Impact End Users' Mental Models. In *2013 IEEE Symposium on Visual Languages and Human Centric Computing*. IEEE, 3–10.
- [13] Marianna Obrist, Rob Comber, Sriram Subramanian, Betina Piqueras-Fiszman, Carlos Velasco, and Charles Spence. 2014. Temporal, Affective, and Embodied Characteristics of Taste Experiences: A Framework for Design. In *Proceedings of the SIGCHI Conference on Human Factors in Computing Systems*. 2853–2862.
- [14] Marco Tulio Ribeiro, Sameer Singh, and Carlos Guestrin. 2016. "Why Should I Trust You?": Explaining the Predictions of Any Classifier. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. 1135–1144.

- [15] Kimiko Ryokai, Elena Durán López, Noura Howell, Jon Gillick, and David Bamman. 2018. Capturing, Representing, and Interacting with Laughter. In *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. 1–12. doi:10.1145/3173574.3173932
- [16] Qian Yang, Aaron Steinfeld, Carolyn Rosé, and John Zimmerman. 2020. Re-examining Whether, Why, and How Human-AI Interaction Is Uniquely Difficult to Design. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems*. 1–13.
- [17] Yanyun Zhou and Chi-Shing Tse. 2020. The Taste of Emotion: Metaphoric Association Between Taste Words and Emotion/Emotion-Laden Words. *Frontiers in Psychology* 11 (2020), 986. doi:10.3389/fpsyg.2020.00986